



Vectors

A VECTOR IS A LINE THAT HAS SIZE (MAGNITUDE) AND DIRECTION.

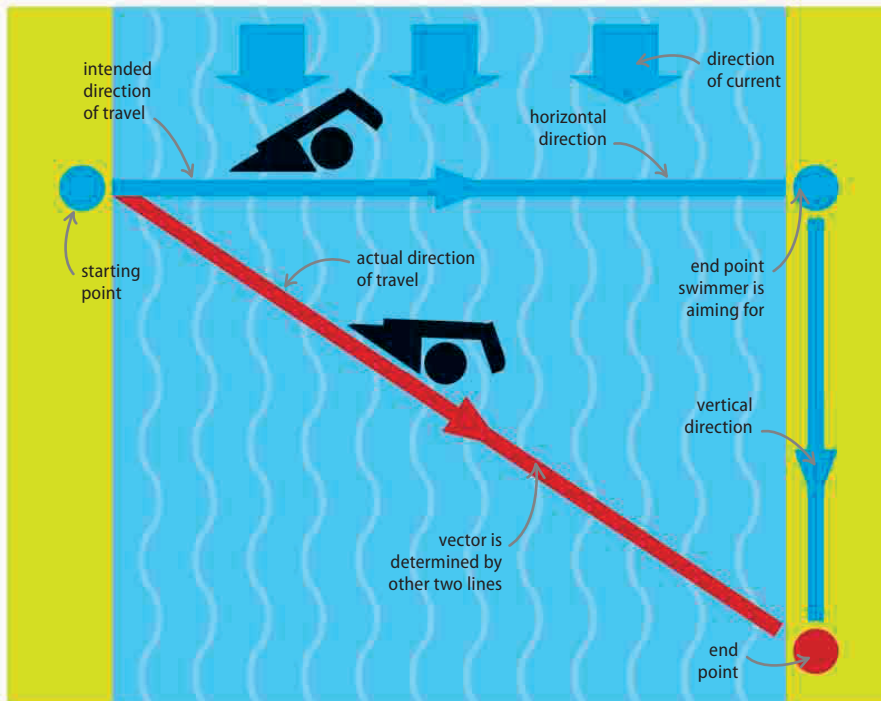
A vector is a way to show a distance in a particular direction. It is often drawn as a line with an arrow on it. The length of the line shows the size of the vector and the arrow gives its direction.

SEE ALSO

◀ 90–93 Coordinates

Translations 98–99 ▶

Pythagorean Theorem 128–129 ▶



What is a vector?

A vector is a distance in a particular direction. Often, a vector is a diagonal distance, and in these cases it forms the diagonal side (hypotenuse) of a right-angled triangle (see pp.128–129). The other sides of the triangle determine the vector's length and direction. In the example on the left, a swimmer's path is a vector. The other two sides of the triangle are the distance across to the opposite shore from the starting point, and the distance down from the end point that the swimmer was aiming for to the actual end point where the swimmer reaches the shore.

◀ Vector of a swimmer

A man sets out to swim to the opposite shore of a river that is 30m wide. A current pushes him as he swims, and he ends up 20m downriver from where he intended. His path is a vector with dimensions 30 across and 20 down.

Expressing vectors

In diagrams, a vector is drawn as a line with an arrow on it, showing its size and direction. There are three different ways of writing vectors using letters and numbers.

$\mathbf{v} =$

A "v" is a general label for a vector, used even when its size is known. It is often used as a label in a diagram.

$\overrightarrow{ab} =$

Another way of representing a vector is by giving its start and end points, with an arrow above them to show direction.

$\begin{pmatrix} 6 \\ 4 \end{pmatrix} =$

The size and direction of the vector can be shown by giving the horizontal units over the vertical units.

